

PROJECT PLANNING & SPRINT EXECUTION

Team ID: SWTID-2026-3238 | Project Name: PocketSmart AI | Track: AI-ML & Generative AI

1. Work Breakdown Structure (WBS)

Sprint	Work Package	Deliverables	Owner
Sprint 1	Core Environment & DB	Virtual env, FastAPI setup, SQLAlchemy schema, JWT auth pipeline	Siddharth, Monish
Sprint 2	Deterministic Planners	Home, Party, and Jewelry budget allocation algorithms and schemas	Sai Prasanna, Monish
Sprint 3	GenAI & Multimodal SDK	Gemini 2.5 Flash prompt engineering, vision client, fallback engine	Siddharth
Sprint 4	UI/UX & Dynamic JS	Jinja2 templates, CSS design system, asynchronous AJAX form submit	Sai Prasanna
Sprint 5	QA, Testing & Docs	Pytest automated test suite, load testing, comprehensive docx report	Thameem Ansari K

2. Risk Management Matrix

Identified Risk	Severity	Mitigation Strategy Implemented
Gemini API Quota Exhaustion or Timeout	High	Built zero-latency deterministic fallback delivering complete plans without API keys.
Third-Party Retail API Deprecation	Medium	Constructed provider-aware search query builders (Amazon/IKEA/Zomato).
Unauthorized History Access	High	Enforced strict JWT token extraction from HTTP-only session cookies with per-user SQLAlchemy filtering.